

GRE over IPsec

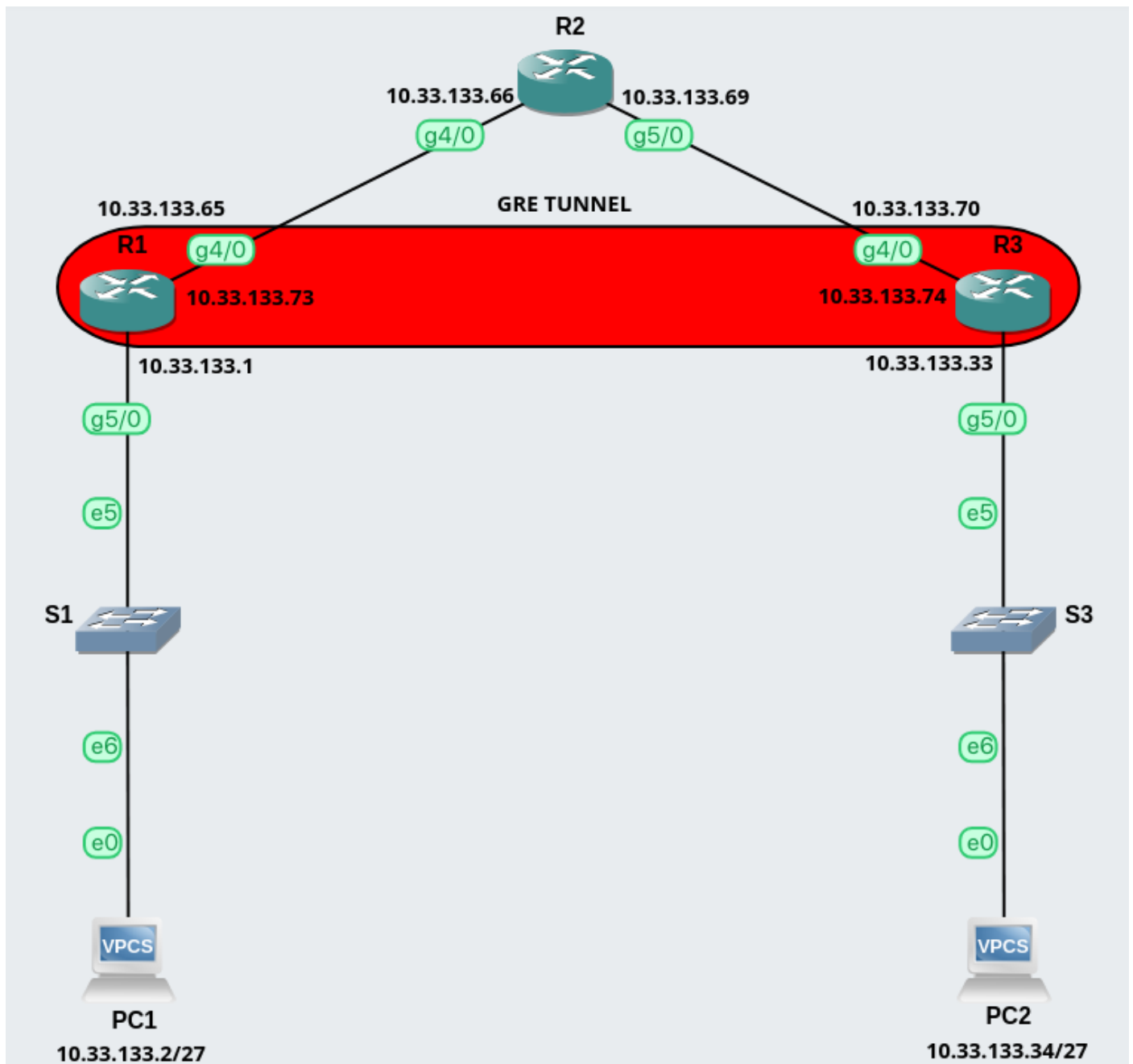
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GROUP 1

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Topology



IP Addressing Plan

a. R1

```
R1#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
FastEthernet0/0    unassigned      YES unset  administratively down down
FastEthernet0/1    unassigned      YES unset  administratively down down
Serial1/0           unassigned      YES unset  administratively down down
Serial1/1           unassigned      YES unset  administratively down down
Serial1/2           unassigned      YES unset  administratively down down
Serial1/3           unassigned      YES unset  administratively down down
Serial1/4           unassigned      YES unset  administratively down down
Serial1/5           unassigned      YES unset  administratively down down
Serial1/6           unassigned      YES unset  administratively down down
Serial1/7           unassigned      YES unset  administratively down down
Serial2/0           unassigned      YES unset  administratively down down
Serial2/1           unassigned      YES unset  administratively down down
Serial2/2           unassigned      YES unset  administratively down down
Serial2/3           unassigned      YES unset  administratively down down
Ethernet3/0         unassigned      YES unset  administratively down down
Ethernet3/1         unassigned      YES unset  administratively down down
Ethernet3/2         unassigned      YES unset  administratively down down
Ethernet3/3         unassigned      YES unset  administratively down down
Ethernet3/4         unassigned      YES unset  administratively down down
Ethernet3/5         unassigned      YES unset  administratively down down
Ethernet3/6         unassigned      YES unset  administratively down down
Ethernet3/7         unassigned      YES unset  administratively down down
GigabitEthernet4/0  10.33.133.65   YES manual up              up
GigabitEthernet5/0  10.33.133.1    YES manual up              up
GigabitEthernet6/0  unassigned      YES unset  administratively down down
Tunnel0            10.33.133.73   YES manual up              up
R1#
```

b. R2

```
R2#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
FastEthernet0/0    unassigned      YES unset  administratively down down
FastEthernet0/1    unassigned      YES unset  administratively down down
Serial1/0           unassigned      YES unset  administratively down down
Serial1/1           unassigned      YES unset  administratively down down
Serial1/2           unassigned      YES unset  administratively down down
Serial1/3           unassigned      YES unset  administratively down down
Serial1/4           unassigned      YES unset  administratively down down
Serial1/5           unassigned      YES unset  administratively down down
Serial1/6           unassigned      YES unset  administratively down down
Serial1/7           unassigned      YES unset  administratively down down
Serial2/0           unassigned      YES unset  administratively down down
Serial2/1           unassigned      YES unset  administratively down down
Serial2/2           unassigned      YES unset  administratively down down
Serial2/3           unassigned      YES unset  administratively down down
Ethernet3/0         unassigned      YES unset  administratively down down
Ethernet3/1         unassigned      YES unset  administratively down down
Ethernet3/2         unassigned      YES unset  administratively down down
Ethernet3/3         unassigned      YES unset  administratively down down
Ethernet3/4         unassigned      YES unset  administratively down down
Ethernet3/5         unassigned      YES unset  administratively down down
Ethernet3/6         unassigned      YES unset  administratively down down
Ethernet3/7         unassigned      YES unset  administratively down down
GigabitEthernet4/0  10.33.133.66   YES manual up              up
GigabitEthernet5/0  10.33.133.69   YES manual up              up
GigabitEthernet6/0  unassigned      YES unset  administratively down down
R2#
```

c. R3

```
R3#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
FastEthernet0/0          unassigned      YES unset  administratively down  down
FastEthernet0/1          unassigned      YES unset  administratively down  down
Serial1/0                unassigned      YES unset  administratively down  down
Serial1/1                unassigned      YES unset  administratively down  down
Serial1/2                unassigned      YES unset  administratively down  down
Serial1/3                unassigned      YES unset  administratively down  down
Serial1/4                unassigned      YES unset  administratively down  down
Serial1/5                unassigned      YES unset  administratively down  down
Serial1/6                unassigned      YES unset  administratively down  down
Serial1/7                unassigned      YES unset  administratively down  down
Serial2/0                unassigned      YES unset  administratively down  down
Serial2/1                unassigned      YES unset  administratively down  down
Serial2/2                unassigned      YES unset  administratively down  down
Serial2/3                unassigned      YES unset  administratively down  down
Ethernet3/0              unassigned      YES unset  administratively down  down
Ethernet3/1              unassigned      YES unset  administratively down  down
Ethernet3/2              unassigned      YES unset  administratively down  down
Ethernet3/3              unassigned      YES unset  administratively down  down
Ethernet3/4              unassigned      YES unset  administratively down  down
Ethernet3/5              unassigned      YES unset  administratively down  down
Ethernet3/6              unassigned      YES unset  administratively down  down
Ethernet3/7              unassigned      YES unset  administratively down  down
GigabitEthernet4/0       10.33.133.70    YES manual up                    up
GigabitEthernet5/0       10.33.133.33    YES manual up                    up
GigabitEthernet6/0       unassigned      YES unset  administratively down  down
Tunnel0                  10.33.133.74    YES manual up                    down
R3#
```

d. PC1

```
PC1> show ip
NAME       : PC1[1]
IP/MASK    : 10.33.133.2/27
GATEWAY    : 10.33.133.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 20024
RHOST:PORT : 127.0.0.1:20025
MTU        : 1500
```

e. PC2

```
PC2> show ip

NAME       : PC2[1]
IP/MASK    : 10.33.133.34/27
GATEWAY    : 10.33.133.33
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 20026
RHOST:PORT : 127.0.0.1:20027
MTU        : 1500
```

GRE over IPsec Configuration

f. R1

```
R1#show run | section interface Tunnel0

*Mar  3 09:17:36.427: %ADJ-5-PARENT: Midchain parent maintenance for IP midchain out of Tunnel0 - looped chain attempting to stackinterface Tunnel0
ip address 10.33.133.73 255.255.255.252
tunnel source 10.33.133.65
tunnel destination 10.33.133.70
R1#show run | section crypto

*Mar  3 09:17:40.791: %TUN-5-RECURDOWN: Tunnel0 temporarily disabled due to recursive routing
*Mar  3 09:17:40.795: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to down
*Mar  3 09:17:40.815: %OSPF-5-ADJCHG: Process 1, Nbr 10.33.133.74 on Tunnel0 from FULL to DOWN, Neighbor Down: Interface down or detachedcrypto isakmp policy 10
encr aes
authentication pre-share
group 2
crypto isakmp key vpn123 address 10.33.133.70
crypto ipsec transform-set TS esp-aes esp-sha-hmac
mode transport
crypto map CMAP 10 ipsec-isakmp
set peer 10.33.133.70
set transform-set TS
match address 100
crypto map CMAP
```

g. R2

```
R3#show run | section interface Tunnel0
interface Tunnel0
ip address 10.33.133.74 255.255.255.252
tunnel source 10.33.133.70
tunnel destination 10.33.133.65
R3#show run | section crypto
crypto isakmp policy 10
encr aes
authentication pre-share
group 2
crypto isakmp key vpn123 address 10.33.133.65
crypto ipsec transform-set TS esp-aes esp-sha-hmac
mode transport
crypto map CMAP 10 ipsec-isakmp
set peer 10.33.133.65
set transform-set TS
match address 100
crypto map CMAP
R3#
```

VPN Verification

```
R3#show interface tunnel0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 10.33.133.74/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 10.33.133.70, destination 10.33.133.65
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:00:43, output 00:00:06, output hang never
  Last clearing of "show interface" counters 00:19:01
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 12
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    160 packets input, 16876 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    219 packets output, 26916 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out
R3#
```

- IKE Phase 1

```
R3#
R3#show crypto isakmp sa
*Mar  3 09:20:33.995: %OSPF-5-ADJCHG: Process 1, Nbr 10.33.133.65 on Tunnel0 from FULL to DOWN, Neighbor Down: Dead timer expired
IPv4 Crypto ISAKMP SA

```

dst	src	state	conn-id	status
10.33.133.70	10.33.133.65	QM_IDLE	1002	ACTIVE
10.33.133.65	10.33.133.70	QM_IDLE	1001	ACTIVE

```
IPv6 Crypto ISAKMP SA
R3#
```

- IKE Phase 2

```
R3#show crypto ipsec sa

interface: GigabitEthernet4/0
  Crypto map tag: CMAP, local addr 10.33.133.70

protected vrf: (none)
local ident (addr/mask/prot/port): (10.33.133.70/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (10.33.133.65/255.255.255.255/47/0)
current_peer 10.33.133.65 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 189, #pkts encrypt: 189, #pkts digest: 189
  #pkts decaps: 175, #pkts decrypt: 175, #pkts verify: 175
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

  local crypto endpt.: 10.33.133.70, remote crypto endpt.: 10.33.133.65
  path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet4/0
  current outbound spi: 0x24F1A45(38738501)
  PFS (Y/N): N, DH group: none

inbound esp sas:
  spi: 0x9731934D(2536608589)
--More--
*Mar  3 09:21:01.503: %OSPF-5-ADJCHG: Process 1, Nbr 10.33.133.65 on Tunnel0 from LOADING to FULL, Lo
  in use settings =(Transport, )
  conn id: 1, flow_id: 1, sibling_flags 80004000, crypto map: CMAP
  sa timing: remaining key lifetime (k/sec): (4608000/2373)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)
spi: 0xEFC0DB0D(4022393613)
  transform: esp-aes esp-sha-hmac ,
  in use settings =(Transport, )
  conn id: 3, flow_id: 3, sibling_flags 80000000, crypto map: CMAP
  sa timing: remaining key lifetime (k/sec): (4274555/2374)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)

inbound ah sas:

inbound pcp sas:

outbound esp sas:
  spi: 0xD6A735AD(3601282477)
  transform: esp-aes esp-sha-hmac ,
```

Connectivity Testing

h. PC1-PC2

```
PC1> ping 10.33.133.34

84 bytes from 10.33.133.34 icmp_seq=1 ttl=61 time=61.470 ms
84 bytes from 10.33.133.34 icmp_seq=2 ttl=61 time=63.807 ms
84 bytes from 10.33.133.34 icmp_seq=3 ttl=61 time=54.527 ms
84 bytes from 10.33.133.34 icmp_seq=4 ttl=61 time=60.925 ms
84 bytes from 10.33.133.34 icmp_seq=5 ttl=61 time=41.157 ms
```

i. PC2-PC1

```
PC2> ping 10.33.133.2

84 bytes from 10.33.133.2 icmp_seq=1 ttl=61 time=130.302 ms
84 bytes from 10.33.133.2 icmp_seq=2 ttl=61 time=53.652 ms
84 bytes from 10.33.133.2 icmp_seq=3 ttl=61 time=54.131 ms
84 bytes from 10.33.133.2 icmp_seq=4 ttl=61 time=42.711 ms
84 bytes from 10.33.133.2 icmp_seq=5 ttl=61 time=52.110 ms
```

Saved Configurations

j. R1

```
R1#show startup-config
```

```
Using 2855 out of 129016 bytes!
```

```
!
```

```
!
```

```
! Last configuration change at 08:59:59 UTC Tue Mar 3 2026
```

```
!
```

```
version 15.2
```

```
service timestamps debug datetime msec
```

```
service timestamps log datetime msec
```

```
!
```

```
hostname R1
```


!

boot-start-marker

boot-end-marker

!

!

!

no aaa new-model

ip cef

!

!

!

!

!

!

no ipv6 cef

!

!

multilink bundle-name authenticated

!

!

crypto isakmp policy 10

encr aes

authentication pre-share

group 2

crypto isakmp key vpn123 address 10.33.133.70

!

!

crypto ipsec transform-set TS esp-aes esp-sha-hmac

mode transport

!

!

!

```
crypto map CMAP 10 ipsec-isakmp
```

```
set peer 10.33.133.70
```

```
set transform-set TS
```

```
match address 100
```

!

!

!

!

!

```
interface Tunnel0
```

!

```
ip address 10.33.133.73 255.255.255.252
```

```
tunnel source 10.33.133.65
```

```
tunnel destination 10.33.133.70
```

```
FastEthernet interface 0/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
speed auto
```

```
duplex auto
```

```
FastEthernet0/1 interface
```

```
no ip address
```

```
shutdown
```

```
speed auto
```

```
duplex auto
```

```
!
```

```
speed auto
```

```
duplex auto
```

```
interface Serial1/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/1
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/2
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/3
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/4
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/5
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/6
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/7
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial2/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial2/1
```

```
interface Serial2/2
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial2/3
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Ethernet3/0
```


!

no ip address

shutdown

duplex full

interface Ethernet3/1

!

no ip address

shutdown

duplex full

interface Ethernet3/2

!

no ip address

shutdown

duplex full

```
interface Ethernet3/3
```

```
!
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
interface Ethernet3/4
```

```
!
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
interface Ethernet3/5
```

```
!
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
interface Ethernet3/6
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
!
```

```
!
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
interface Ethernet3/7
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
i
```

```
interface GigabitEthernet4/0
```

!

```
ip address 10.33.133.65 255.255.255.252
```

```
negotiation auto
```

```
crypto map CPAR
```

```
interface GigabitEthernet5/0
```

i

```
ip address 10.33.133.1 255.255.255.224
```

```
negotiation auto
```

```
interface GigabitEthernet6/0
```

!

```
no ip address
```

```
shutdown
```

```
negotiation auto
```

```
router ospf 1
```

!

```
network 10.33.133.0 0.0.0.255 area 0
```

```
ip forward-protocol nd
```

!

!

```
no ip http server
```

```
no ip http secure-server
```

```
ip route 0.0.0.0 0.0.0.0 10.33.133.66
```

!

```
access-list 100 permit gre host 10.33.133.65 host 10.33.133.70
```

!

!

```
control-plane
```

!

!

```
line con 0
```

```
stopbits 1
```

```
line aux 0
```

```
stopbits 1
```

```
line vty 0 4
```

```
!
```

```
login
```

```
i
```

```
end
```

k. R2

```
R2#show start
```

```
R2#show startup-config
```

```
Using 2373 out of 129016 bytes!
```

```
!
```

!

! Last configuration change at 09:00:13 UTC Tue Mar 3 2026

!

version 15.2

service timestamps debug datetime msec

service timestamps log datetime msec

!

hostname R2

!

boot-start-marker

boot-end-marker

!

!

!

```
no aaa new-model
```

```
ip cef
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
no ipv6 cef
```

```
!
```

```
!
```

```
multilink bundle-name authenticated
```

```
!
```

```
!
```

```
boot-end-marker
```



```
no aaa new-model
```

```
ip cef
```

```
!
```

```
!
```

```
no ipv6 cef
```

```
!
```

```
i
```

```
multilink bundle-name authenticated
```

```
!
```

```
FastEthernet interface 0/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
speed auto
```

duplex auto

FastEthernet0/1 interface

!

no ip address

shutdown

speed auto

duplex auto

!

shutdown

speed auto

duplex auto

interface Serial1/0

!

no ip address

shutdown

```
serial restart-delay 0
```

```
interface Serial1/1
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/2
```

```
i
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/3
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/4
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/5
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/6
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/7
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial2/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
!
```

```
interface Serial2/1
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial2/2
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial2/3
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Ethernet3/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
interface Ethernet3/1
```

```
!
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
interface Ethernet3/2
```

```
!
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
interface Ethernet3/3
```

!

no ip address

shutdown

duplex full

interface Ethernet3/4

!

no ip address

shutdown

duplex full

interface Ethernet3/5

no ip address

shutdown

duplex full

interface Ethernet3/6

i


```
no ip address
```

```
shutdown
```

```
duplex full
```

```
interface Ethernet3/7
```

```
!
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
interface GigabitEthernet4/0
```

```
!
```

```
ip address 10.33.133.66 255.255.255.252
```

```
negotiation auto
```

```
interface GigabitEthernet5/0
```

```
!
```

```
ip address 10.33.133.69 255.255.255.252
```

```
negotiation auto
```

```
interface GigabitEthernet6/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
negotiation auto
```

```
ip forward-protocol nd
```

```
!
```

```
!
```

```
no ip http server
```

```
no ip http secure-server
```

```
ip route 10.33.133.0 255.255.255.224 10.33.133.65
```

```
ip route 10.33.133.32 255.255.255.224 10.33.133.70
```

```
!
```

!

!

!

control-plane

!

!

line con 0

stopbits 1

line aux 0

stopbits 1

line vty 0 4

!

login

i

```
end
```

```
R2#
```

```
l. R3
```

```
R3#show startup-config
```

```
Using 2856 out of 129016 bytes!
```

```
!
```

```
!
```

```
! Last configuration change at 09:00:27 UTC Tue Mar 3 2026
```

```
!
```

```
version 15.2
```

```
service timestamps debug datetime msec
```

```
service timestamps log datetime msec
```

```
!
```

```
hostname R3
```

```
!
```

```
boot-start-marker
```

```
boot-end-marker
```

```
!
```

```
!
```

```
!
```

```
no aaa new-model
```

```
ip cef
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
no ipv6 cef
```

!

!

multilink bundle-name authenticated

!

!

!

crypto isakmp policy 10

encr aes

authentication pre-share

group 2

crypto isakmp key vpn123 address 10.33.133.65

!

!

crypto ipsec transform-set TS esp-aes esp-sha-hmac

mode transport

!

!

!

```
crypto map CMAP 10 ipsec-isakmp
```

```
set peer 10.33.133.65
```

```
set transform-set TS
```

```
match address 100
```

!

!

!

!

!

```
interface Tunnel0
```

!

```
ip address 10.33.133.74 255.255.255.252
```

```
tunnel source 10.33.133.70
```

```
tunnel destination 10.33.133.65
```

```
FastEthernet interface 0/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
speed auto
```

```
duplex auto
```

```
FastEthernet0/1 interface
```

```
!
```

```
no ip address
```

```
shutdown
```

```
speed auto
```

```
duplex auto
```



```
interface Serial1/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
!
```

```
interface Serial1/1
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/2
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/3
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/4
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/5
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/6
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial1/7
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial2/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial2/1
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
!
```

```
interface Serial2/2
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Serial2/3
```

```
!
```

```
no ip address
```

```
shutdown
```

```
serial restart-delay 0
```

```
interface Ethernet3/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
interface Ethernet3/1
```

```
!
```

```
no ip address
```

```
shutdown
```

```
duplex full
```

```
interface Ethernet3/2
```

!

no ip address

shutdown

duplex full

interface Ethernet3/3

!

no ip address

shutdown

duplex full

interface Ethernet3/4

!

no ip address

shutdown

duplex full

interface Ethernet3/5

!

no ip address

shutdown

duplex full

interface Ethernet3/6

no ip address

shutdown

duplex full

!

!

interface Ethernet3/7

!

no ip address

shutdown

```
duplex full
```

```
interface GigabitEthernet4/0
```

```
ip address 10.33.133.70 255.255.255.252
```

```
negotiation auto
```

```
crypto map CPAR
```

```
interface GigabitEthernet5/0
```

```
!
```

```
ip address 10.33.133.33 255.255.255.224
```

```
negotiation auto
```

```
interface GigabitEthernet 6/0
```

```
!
```

```
no ip address
```

```
shutdown
```

```
negotiation auto
```

```
router ospf 1
```


!

```
network 10.33.133.0 0.0.0.255 area 0
```

```
ip forward-protocol nd
```

!

!

```
no ip http server
```

```
no ip http secure-server
```

```
ip route 0.0.0.0 0.0.0.0 10.33.133.69
```

i

```
access-list 100 permit gre host 10.33.133.70 host 10.33.133.65
```

!

!

i

```
control-plane
```

!

!

line con 0

stopbits 1

line aux 0

stopbits 1

line vty 04

login

!

!

end

R3#